

Lesson 3 - Practical Application Guide

□ How to Run the Project

```
cd D:\learnlaravel2025\lessons\lesson-03\practice-app
php artisan serve
```

Server will run on: `http://localhost:8000`

□ Available Routes

1. Basic Pages

- **GET** / - Home page
- **GET** /about - About us
- **GET** /contact - Contact us

2. Products - Resource Controller

- **GET** /products - Products list
- **GET** /products/create - Add product form
- **POST** /products - Save new product
- **GET** /products/{id} - Show single product
- **GET** /products/{id}/edit - Edit product form
- **PUT/PATCH** /products/{id} - Update product
- **DELETE** /products/{id} - Delete product

3. Posts - Resource Controller

- **GET** /posts - Posts list
- **GET** /posts/create - Add post form
- **POST** /posts - Save new post
- **GET** /posts/{id} - Show single post
- **GET** /posts/{id}/edit - Edit post form

- **PUT/PATCH** /posts/{id} - Update post
- **DELETE** /posts/{id} - Delete post

4. Dashboard

- **GET** /dashboard - Dashboard (Single Action Controller)

5. Users

- **POST** /users - Create new user (with validation)
 - **PUT** /users/{id} - Update user (with validation)
-

📁 Implemented Controllers

1. PageController - Basic Pages

```
php artisan make:controller PageController
```

```
class PageController extends Controller
{
    public function home()
    {
        return view('pages.home');
    }

    public function about()
    {
        return view('pages.about');
    }

    public function contact()
    {
        return view('pages.contact');
    }
}
```

Routes:

```
Route::get('/', [PageController::class, 'home'])->name('home');
```

```
Route::get('/about', [PageController::class, 'about'])->name('about');
Route::get('/contact', [PageController::class, 'contact'])->name('contact');
```

2. ProductController - Resource Controller

```
php artisan make:controller ProductController --resource
```

```
class ProductController extends Controller
{
    // Display all products
    public function index()
    {
        $products = [
            ['id' => 1, 'name' => 'HP Laptop', 'price' => 5000, 'stock' => 10],
            ['id' => 2, 'name' => 'iPhone', 'price' => 4000, 'stock' => 15],
            ['id' => 3, 'name' => 'Samsung Tablet', 'price' => 2000, 'stock' => 8],
        ];

        return view('products.index', compact('products'));
    }

    // Show form to create new product
    public function create()
    {
        return view('products.create');
    }

    // Save new product
    public function store(Request $request)
    {
        $validated = $request->validate([
            'name' => 'required|max:255',
            'price' => 'required|numeric|min:0',
            'stock' => 'required|integer|min:0',
        ]);

        // In real app: Product::create($validated);

        return redirect()->route('products.index')
            ->with('success', 'Product added successfully');
    }

    // Display single product
}
```

```

public function show($id)
{
    $products = [
        1 => ['id' => 1, 'name' => 'HP Laptop', 'price' => 5000, 'stock' =>
10],
        2 => ['id' => 2, 'name' => 'iPhone', 'price' => 4000, 'stock' => 15],
        3 => ['id' => 3, 'name' => 'Samsung Tablet', 'price' => 2000, 'stock'
=> 8],
    ];

    $product = $products[$id] ?? abort(404);

    return view('products.show', compact('product'));
}

// Show form to edit product
public function edit($id)
{
    $products = [
        1 => ['id' => 1, 'name' => 'HP Laptop', 'price' => 5000, 'stock' =>
10],
        2 => ['id' => 2, 'name' => 'iPhone', 'price' => 4000, 'stock' => 15],
        3 => ['id' => 3, 'name' => 'Samsung Tablet', 'price' => 2000, 'stock'
=> 8],
    ];

    $product = $products[$id] ?? abort(404);

    return view('products.edit', compact('product'));
}

// Update product
public function update(Request $request, $id)
{
    $validated = $request->validate([
        'name' => 'required|max:255',
        'price' => 'required|numeric|min:0',
        'stock' => 'required|integer|min:0',
    ]);

    // In real app: $product->update($validated);

    return redirect()->route('products.show', $id)
        ->with('success', 'Product updated successfully');
}

// Delete product
public function destroy($id)

```

```

{
    // In real app: $product->delete();

    return redirect()->route('products.index')
        ->with('success', 'Product deleted successfully');
}
}

```

Routes:

```
Route::resource('products', ProductController::class);
```

3. PostController - Resource Controller

```
php artisan make:controller PostController --resource
```

```

class PostController extends Controller
{
    public function index()
    {
        $posts = [
            ['id' => 1, 'title' => 'Introduction to Laravel', 'author' => 'Ahmed
Mohammed', 'date' => '2024-01-15'],
            ['id' => 2, 'title' => 'Learning Controllers', 'author' => 'Mohammed
Ali', 'date' => '2024-01-20'],
            ['id' => 3, 'title' => 'MVC Pattern', 'author' => 'Sara Ahmed', 'date'
=> '2024-01-25'],
        ];

        return view('posts.index', compact('posts'));
    }

    public function create()
    {
        return view('posts.create');
    }

    public function store(Request $request)
    {
        $validated = $request->validate([
            'title' => 'required|max:255',
            'content' => 'required|min:10',

```

```

        'author' => 'required|max:100',
    ]));

    return redirect()->route('posts.index')
        ->with('success', 'Post created successfully');
}

public function show($id)
{
    $posts = [
        1 => [
            'id' => 1,
            'title' => 'Introduction to Laravel',
            'content' => 'Laravel is a modern PHP framework...',
            'author' => 'Ahmed Mohammed',
            'date' => '2024-01-15'
        ],
        2 => [
            'id' => 2,
            'title' => 'Learning Controllers',
            'content' => 'Controllers are the layer that connects...',
            'author' => 'Mohammed Ali',
            'date' => '2024-01-20'
        ],
    ];

    $post = $posts[$id] ?? abort(404);

    return view('posts.show', compact('post'));
}

public function edit($id)
{
    $posts = [
        1 => ['id' => 1, 'title' => 'Introduction to Laravel', 'content' =>
'Laravel is...', 'author' => 'Ahmed'],
        2 => ['id' => 2, 'title' => 'Learning Controllers', 'content' =>
'Controllers are...', 'author' => 'Mohammed'],
    ];

    $post = $posts[$id] ?? abort(404);

    return view('posts.edit', compact('post'));
}

public function update(Request $request, $id)
{
    $validated = $request->validate([

```

```

        'title' => 'required|max:255',
        'content' => 'required|min:10',
        'author' => 'required|max:100',
    ]);

    return redirect()->route('posts.show', $id)
        ->with('success', 'Post updated successfully');
}

public function destroy($id)
{
    return redirect()->route('posts.index')
        ->with('success', 'Post deleted successfully');
}
}

```

Routes:

```
Route::resource('posts', PostController::class);
```

4. ShowDashboardController - Single Action

```
php artisan make:controller ShowDashboardController --invokable
```

```

class ShowDashboardController extends Controller
{
    public function __invoke()
    {
        $stats = [
            'users' => 1250,
            'products' => 85,
            'posts' => 342,
            'revenue' => 125000,
            'orders' => 567,
            'visitors' => 8932
        ];

        return view('dashboard', compact('stats'));
    }
}

```

Route:

```
Route::get('/dashboard', ShowDashboardController::class)
    ->name('dashboard');
```

5. UserController - Dependency Injection & Validation

```
php artisan make:controller UserController
```

```
use Illuminate\Http\Request;

class UserController extends Controller
{
    // Create new user with validation
    public function store(Request $request)
    {
        $validated = $request->validate([
            'name' => 'required|max:255|min:3',
            'email' => 'required|email|unique:users',
            'password' => 'required|min:8|confirmed',
            'phone' => 'nullable|regex:/^[0-9]{10}$/ ',
        ], [
            'name.required' => 'Name is required',
            'email.required' => 'Email is required',
            'email.email' => 'Invalid email address',
            'password.required' => 'Password is required',
            'password.min' => 'Password must be at least 8 characters',
        ]);

        // User::create($validated);

        return redirect()->route('home')
            ->with('success', 'User created successfully');
    }

    // Update user
    public function update(Request $request, $id)
    {
        $validated = $request->validate([
            'name' => 'required|max:255|min:3',
            'email' => 'required|email|unique:users,email,' . $id,
            'phone' => 'nullable|regex:/^[0-9]{10}$/ ',
        ];
```



```

    });

    // $user->update($validated);

    return redirect()->route('home')
        ->with('success', 'User updated successfully');
}
}

```

Routes:

```

Route::post('/users', [UserController::class, 'store'])->name('users.store');
Route::put('/users/{id}', [UserController::class, 'update'])->name('users.update');

```

□ What We Learned

1. Controller Types

- □ Simple Controller (PageController)
- □ Resource Controller (ProductController, PostController)
- □ Single Action Controller (ShowDashboardController)
- □ Controller with Dependency Injection (UserController)

2. Resource Controller Methods

- `index()` - Display list
- `create()` - Show creation form
- `store()` - Save data
- `show($id)` - Display single item
- `edit($id)` - Show edit form
- `update($id)` - Update data
- `destroy($id)` - Delete item

3. Data Validation

- Using `validate()` in Controller
- Custom error messages

- Different validation rules

4. Responses

- `return view()` - Display page
 - `return redirect()` - Redirect
 - `->with('success', 'message')` - Flash messages
-

□ Useful Commands

```
# Display all routes
php artisan route:list

# Display products routes only
php artisan route:list --name=products

# Display posts routes only
php artisan route:list --name=posts

# Create new controller
php artisan make:controller ControllerName --resource

# Detailed routes view
php artisan route:list -v
```

□ Testing Routes

Testing Products:

1. □ `http://localhost:8000/products` - Products list
2. □ `http://localhost:8000/products/1` - Show product #1
3. □ `http://localhost:8000/products/create` - Add product form
4. □ `http://localhost:8000/products/1/edit` - Edit product form

Testing Posts:

1. `http://localhost:8000/posts` - Posts list
2. `http://localhost:8000/posts/1` - Show post #1
3. `http://localhost:8000/posts/create` - Add post form
4. `http://localhost:8000/posts/1/edit` - Edit post form

Testing Pages:

1. `http://localhost:8000/` - Home page
 2. `http://localhost:8000/about` - About us
 3. `http://localhost:8000/contact` - Contact us
 4. `http://localhost:8000/dashboard` - Dashboard
-

☐ Tips

1. **Use `php artisan route:list`** to display all routes
 2. **Resource Controllers** save your time - use them!
 3. **Single Action Controllers** are useful for complex pages
 4. **Validation** is very important - don't forget it
 5. **Test every route** to ensure it works
-

☐ Additional Exercises

Exercise 1: Create Category Controller

Create a Resource Controller for categories with all 7 methods.

Exercise 2: Add Middleware

Add authentication middleware to `create`, `store`, `edit`, `update`, `destroy`.

Exercise 3: Form Request

Create a Form Request for validation instead of putting it in Controller.

```
php artisan make:request StoreProductRequest
```

📌 Next Step

After completing this lesson, you're now ready for:

Lesson 4: Blade Templates and User Interfaces

- Blade templating engine
 - Directives
 - Components
 - Layouts
-

Happy Learning! 📖